Priyanshu Soni

PhD Researcher — Computational Biomechanics & Biofluid Dynamics

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Research Profile

I am a doctoral researcher in computational and experimental biofluid mechanics at IIT BHU Varanasi, working at the intersection of **computational fluid dynamics (CFD), cardiovascular biomechanics, fluid–structure interaction (FSI), and particle image velocimetry (PIV)-based experimental validation**. My dissertation develops patient-specific, image-driven numerical frameworks for analysing blood flow dynamics under pathological conditions, with direct applications to cardiovascular disease assessment and medical device development. Thesis defence is anticipated between **July and September 2026**.

Supervisors:

- Prof. Sanjay Kumar Rai, School of Biomedical Engineering, IIT BHU Varanasi
  skrai.bme@iitbhu.ac.in | `https://www.iitbhu.ac.in/dept/bme/people/skraibme`
- Prof. B.V. Rathish Kumar, Dept. of Mathematics & Statistics, IIT Kanpur
  bvrk@iitk.ac.in | `https://www.iitk.ac.in/rathish-kumar-b-v`

Research Interests

- Computational Fluid Dynamics (CFD) for cardiovascular and arterial flows
- Fluid–Structure Interaction in arterial pathologies (stenosis, aneurysm)
- Patient-specific modelling from CT/MRI clinical imaging data
- Experimental biofluid mechanics: PIV, flow loop design & validation
- Reduced-order (0D–3D coupled) haemodynamic modelling
- Physics-Informed Neural Networks (PINNs) for cardiovascular flows
- Topology optimisation for biomedical implants and devices

Education

Doctor of Philosophy (PhD) — Cardiovascular Biomechanics *Expected Jul–Sep 2026*
School of Biomedical Engineering, IIT BHU Varanasi, India
Thesis: Numerical and Experimental Analysis of Cardio-Arterial Blood Flow Dynamics for Disease Assessment
Status: Thesis writing; final defence stage.

Master of Technology (M.Tech) — Design Engineering (Mechanical) *2022*
National Institute of Technology (NIT) Hamirpur, India — CGPA: 8.73/10

Bachelor of Technology (B.Tech) — Mechanical Engineering *2019*
Rajasthan Technical University (Govt. Engineering College, Ajmer), India

Research Experience

Doctoral Researcher *IIT BHU Varanasi* 2022–Present

- Developed high-fidelity, patient-specific CFD models of pulsatile flow in the left ventricle and complex arterial networks using clinical CT and MRI data (ANSYS Fluent, OpenFOAM, COMSOL).
- Performed fully coupled FSI simulations incorporating arterial wall viscoelasticity to study proximity effects and multi-pathology haemodynamics (stenosis–aneurysm interactions).

- Built 0D–3D open-loop coupled models for coronary and aorto-femoral systems, significantly reducing computational cost while preserving physiological accuracy.
- Designed, fabricated, and commissioned a pulsatile biofluid flow loop; conducted PIV experiments on transparent arterial phantoms for rigorous numerical validation.
- Quantified advanced haemodynamic risk indices — wall shear stress (WSS), oscillatory shear index (OSI), relative residence time (RRT), endothelial cell activation potential (ECAP), HOLMES — using in-house Python and MATLAB post-processing pipelines.
- Fabricated transparent arterial phantoms (lost-core casting + additive manufacturing) for in-vitro experimental studies.
- Investigated intraventricular vortex formation, evolution, and flow organisation under physiological and pathological conditions.

Junior Research Fellow (ICMR-funded Project) *IIT BHU Varanasi* Jul–Dec 2022

- Contributed to development of a computational cardiac model for non-invasive prediction of heart failure via medical imaging and CFD (Project Grant: INR 52,31,900; PI: Prof. S.K. Rai).

Design Engineering Intern *Mechaton Engineering Pvt. Ltd* Jun–Aug 2021

- Design optimisation of multi-plunger positive displacement pumps (SolidWorks, ANSYS).

Testing Engineering Intern *National Engineering Industries (NEI) Ltd, Jaipur* Mar–Jul 2018

- Machine testing, bearing track roundness analysis, time-study and maintenance management in R&D division.

Technical Expertise

CFD & Simulation	ANSYS Fluent, OpenFOAM, COMSOL Multiphysics, ParaView
Multiphysics	FSI, Multiphase Flow, Turbulence Modelling (k-ω SST, LES), PINNs
Programming	Python, MATLAB, C++ (basic)
CAD / Design	SolidWorks, Fusion 360, AutoCAD
Experimental	Particle Image Velocimetry (PIV), flow loop design & commissioning
Medical Imaging	CT/MRI segmentation, simulation-ready model development (Mimics, 3-Matic)
Languages	English (Professional), Hindi (Native)

Publications

Peer-Reviewed Journal Articles

7. Soni, P., Bhattacharyya, A., Usmani, A.Y., Rai, S.K., & Kumar, B.V.R. (2026). Rortex Pathway to Vortex Analysis in Patient-Specific Left Ventricle Models. *Computer Methods and Programs in Biomedicine* (*Under Review*).

6. Soni, P., Kumar, S., et al. (2026). Numerical modelling of pulsatile blood flow dynamics in three-dimensional human left ventricle models. *Journal of Flow Visualization and Image Processing* DOI: 10.1615/JFlowVisImageProc.2026062466.

5. Bhattacharyya, A., Soni, P., Shrivastava, P., et al. (2026). Structural and functional optimisation of bio-ceramics, metallic oxides and nitrides coatings on topology-optimised mandibular canine implant for enhanced biomechanical responses. *Journal of the Australian Ceramic Society*. doi:10.1007/s41779-026-01355-8

4. Yenurkar, D., Choudhary, A., ... Soni, P., et al. (2025). *ACS Applied Materials & Interfaces*, 17(19), 28395–28410. doi:10.1021/acsami.5c04112

3. Soni, P., Bhattacharyya, A., Usmani, A.Y., Rai, S.K., & Kumar, B.V.R. (2025). Pulsatile flow dynamics in an artery with multiple pathologies: A fluid–structure interaction study. *Physics of*

Fluids, 37(7), 071913. doi:10.1063/5.0263599

2. Soni, P., Kumar, S., Kumar, B.V.R., et al. (2024). A comprehensive review on CFD simulations of left ventricle hemodynamics: numerical methods, experimental validation techniques, and emerging trends. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 46, 301. doi:10.1007/s40430-024-04875-1

1. Soni, P., Shrivastava, P., & Rai, S.K. (2023). Development of reduced-volume endosseous cuspid tooth implant using topology optimisation by SIMP technique for improved osseointegration. *Computer Methods in Biomechanics and Biomedical Engineering*. doi:10.1080/1025584

Book Chapter

1. Soni, P., & Shrivastava, P. (2021). Topology optimisation techniques employed for additively manufactured medical implants. In *Evolutionary Optimization of Material Removal Processes*. CRC Press (Taylor & Francis Group). doi:10.1201/9781003408345

Conference Presentations

4. **Soni, P.**, Bhattacharyya, A., Usmani, A.Y., Rai, S.K., & Kumar, B.V.R. (2025). Pulsatile flow dynamics in an artery with multiple pathologies: A FSI study. *ACMB–JSMB 2025*, Kyoto, Japan, 7–11 July 2025. *[International – Travel Grant recipient]*

3. **Soni, P.**, Bhattacharyya, A., Kumar, B.V.R., Rai, S.K., & Usmani, A.Y. (2024). In-silico investigation of a generalised stenotic–aneurysmal artery: A FSI approach. *ICRAFMN 2024*, Manipal University Jaipur & NIT Uttarakhand, 10–12 July 2024.

2. **Soni, P.**, Kumar, B.V.R., & Rai, S.K. (2023). Transient blood flow dynamics in 3D ellipsoidal models. *ICRAFMN 2023*, Manipal Institute of Technology & NIT Uttarakhand, 12–14 July 2023.

1. **Soni, P.**, & Shrivastava, P. (2021). Design and finite element analysis of endosseous canine implant. *IPDIMS 2021 (3rd Innovative Product Design and Intelligent Manufacturing System)*, Virtual, NIT Rourkela, 30 December 2021.

Grants, Fellowships & Awards

- **Travel Grant** - Council of Scientific and Industrial Research (CSIR) India to present work in WCCM-ECCOMAS 2026 Germany
- **Travel Grant** — Society of Mathematical Biology (SMB), for ACMB–JSMB 2025, Kyoto, Japan (2025)
- **PhD Research Fellowship** — Government of India (2022–Present)
- **ICMR Project Contributor** — INR 52,31,900 grant for cardiac modelling (2022)

Teaching Experience

Teaching Assistant *IIT BHU Varanasi* 2022–Present

- **MED-115: Engineering Graphics** (Undergraduate Laboratory, Mechanical Engineering) — Demonstrated laboratory experiments, assessed reports.
- **BM-504: Advanced Biomechanics** (Postgraduate, Biomedical Engineering) — Course design, quiz preparation, student mentoring.

Professional Memberships

- American Society of Mechanical Engineers (ASME)
- Society for Mathematical Biology (SMB)
- VPH (Virtual Physiological Huma) the society of in-silico medicine
- American Physical Society (APS)

Professional Development & Certifications

- **Analysis Engineer Training Programme** (Udemy) — Static structural, Thermal, Dynamic, and Topology Optimisation using ANSYS
- **SolidWorks 2021 Essential Training** (LinkedIn Learning) — 3D modelling, Advanced assembly, Engineering drawing
- CMIA E-Workshop, NIT Hamirpur (2021)
- Workshop: Technology, Innovation and Challenges in Rapid Product Manufacturing, NIT Hamirpur (2021)

References

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Prof. B.V. Rathish Kumar
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